ENGINEER/RESEARCHER IN OFFENSIVE CYBERSECURITY

As a cybersecurity engineer, I conduct research and develop tools for offensive cybersecurity.

MAXIMILIEN CHAUX
27 years old (20/04/1999)
+33 6 52 30 11 03
maximilienchaux@hotmail.fr
GitHub Website

PROFESSIONAL EXPERIENCES

Research Engineer in offensive cybersecurity 2025-2026, Freelance

- Development of offensive security research projects
- Development of a website with a blog section to showcase my work

Cybersecurity engineer 2020-2025 (3 years on a work-study basis), Neverhack

- Head of Cybersecurity offensive strategy for an automated Red Team software product named Kraken (C&C, beacon) developed in Rust and C
- Development of an executable (PE, C#, BOF) loader on Windows in C, shellcode compatible
- Constant analysis of new attack vectors, mainly on Windows (Process injection, API Hooking/Unhooking, Direct Syscalls, Sleep Obfuscation, Call Stack Spoofing)

PROJECTS

Development of a custom Shellcode Template in Rust

- Complete Rust compilation and syntax workflow that generates shellcode-compatible code
- Development of macros that make using the template easier and more user-friendly
- Development of an LLVM Pass script to fix vtable generation and avoid relocations

Development of a post-quantum Blockchain in Rust

- State-of-the-art research on Blockchain and post-quantum cryptographic algorithms
- Integration of the Falcon-512 post-quantum cryptographic algorithm on the NEAR blockchain using the Rust language

Kernel rootkit development for Windows in C/C#

- Development of a minifilter driver for file dissimulation
- Research for a kernel system for process dissimulation
- Development of a Command & Control server
- Development of a stealth backdoor

Expression swap software development in Python

- Development of a face recognition system using Python
- Handling and transposition of facial expressions using landmarks and Delaunay triangulation

ACQUIRED SKILLS

Proficiency in system development: Rust, C, C++, Python

Mastery of Git, Gitlab, CI/CD tools and Software architecture

In-depth knowledge of the inner workings of Windows from an offensive point of view

Reverse engineering, malware analysis, debugging, Sysinternals

Engineering research methodology, specifically on Windows evasion techniques

Advanced level in English: 990/990 in TOEIC test (2022)

EDUCATION

Cybersecurity engineering school on a work-study basis | 2020 – 2023
ENSIBS
IS security, network security, penetration testing, cybersecurity management, cryptographic protocols

Bachelor of Computer Science | 2019 - 2020
Lyon 1 University, including 4 months of study in Vietnam (shortened due to COVID-19)
Strengthening of Academic technological knowledge

Academic technological diploma in computer science | 2017 - 2019
Lyon 1 University: Algorithms, administration of networks, databases, and systems